Survival Selection Grows a Private Kin Code, Not a Public Language

Chunxuan Yang

Sogang University, Seoul, Republic of Korea
yang2023@sogang.ac.kr yangchunxuan1@gmail.com

Abstract

Can a decodable signalling code *grow* from random weights under survival selection alone, and is what grows public or private to kin? In a minimal hand-coded 2-D lethal-vent world, tiny tied code-table agents evolve a decodable channel. Against an architecture-matched random-fitness control (differing only in the reproduction-fitness signal), selection *sharpens* rather than originates it: clonal descent alone reaches mutual intelligibility (MII) 0.1174 ($\sim$73$\times$ the $1/625$ floor), and survival selection adds a paired win-margin of $+0.0548$ (95% CI $[0.0234, 0.0849]$, n=48) at gen 28, a margin a post-hoc probe shows is a *transient acceleration*. The code stays private to kin (dominant-lineage share 99.1%, N_{eff} 1.02): even under actively sustained diversity, and even when survival is made to depend on decoding non-kin, no public cross-lineage code bootstraps, a *pressure-vs-survival tension*. We also contribute a bidirectional self-audit that catches deflation as readily as inflation.

Submission type: **Late Breaking Abstracts**

Code & verdict artifacts: `https://github.com/yangchunxuan/grown-grounded-language`; every result regenerates from committed, seeded verdict JSONs.

The question and the world

Where does a shared signalling code come from? Most computational accounts *train* a channel in (gradient descent on a task with a differentiable channel; Foerster et al., 2016; Lazaridou et al., 2017) or *design* one in (an imposed iterated-learning transmission bottleneck; Kirby et al., 2008; Ren et al., 2020). We take up the prior question: can a decodable code *grow* from random weights under **survival selection alone**, and if it does, is it a **public** code or a code private to **kin**?

The substrate is deliberately minimal and entirely hand-coded: a 48×48 lethal-vent world in which agents metabolise, can die, and reproduce, and in which a single tied `UnifiedIO` code-table ($4 \times 5 \times 6$, $\mathcal{N}(0,1)$ init) is the only communicative apparatus. There is no gradient on communication, no oracle (agents never read world truth), and no borrowed weights. We measure mutual intelligibility (MII) behaviourally: route a referent through one agent's `emit` into another's `decode` and score exact-tuple recovery against a chance floor of $1/625 \approx 0.0016$. Within-founder (WF) and cross-founder (CF) MII separate a *private kin code* (WF high, CF at chance) from a *public* one (CF also above chance).

Co-evolution is real, but decomposes and stays kin-private

The control is not communication-ablated: both arms carry the full code-table and run the identical loop, and the only manipulated variable is the reproduction-fitness signal (survival vs. uniform-random, with the reproduction RNG byte-matched). This isolates selection from *clonal descent-convergence*. The random-fitness arm also fixates to $\sim$one lineage, so its MII of **0.1174** ($\sim$73$\times$ floor) is the descent baseline, not zero. At gen 28 the survival arm reaches **0.17219**, a paired win-margin of $+0.0548$**, 95% CI** $[0.0234, 0.0849]$ (n=48; Figure 1, panel a). Selection therefore *sharpens* a largely descent-determined channel rather than originating it.

A post-hoc probe (n=24; gen 28/56/100/150) shows this is a *transient acceleration*: the survival-arm MII holds at $\sim$0.16–0.17 while the control's rises ($0.117 \rightarrow 0.138$) as clonal fixation accumulates decodability, so the paired margin decays from $+0.0569$ (CI excludes zero) to a CI including zero by gen 100. The headline is therefore scoped to 28 generations.

The code is *not public*. The survival arm fixates to a single founder lineage (dominant-lineage share **99.1%**, N_{eff} **1.02**, 47/48 seeds single-lineage), so WF runs $\approx$0.16 ($\sim$100$\times$ chance) while CF sits at the floor. Almost every receiver an agent meets is kin, so a private code is selectively cheap (Hamilton, 1964).

© 2026 Chunxuan Yang. Published under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

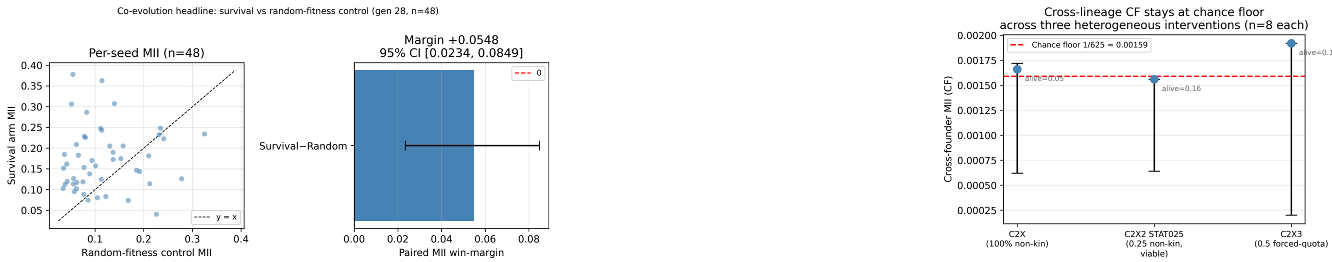

Figure 1: **A channel grows, but only kin can read it.** **(a)** Co-evolution headline (n=48, gen 28): the survival arm beats the architecture-matched random-fitness control by a paired MII win-margin $+0.0548$ (95% CI $[0.0234, 0.0849]$); a post-hoc probe shows this is a transient acceleration. **(b)** Cross-founder MII stays at the chance floor (≈ 0.0016) across three cross-lineage interventions (n=8); where pressure bites, survival crashes. No public code bootstraps. Sources: committed verdict JSONs.

The wall: no public code bootstraps

Is kin-privacy merely an artifact of fast founder collapse? We sustained genuine multi-lineage coexistence by niching (tournament selection plus lineage fitness-sharing) at pop=96: the soft96 cell holds N_{eff} 4.10 over 20+ coexisting generations, yet CF stays at the floor (CF 0.00153 vs. floor 0.0016; the CF−floor CI straddles zero). Maintaining diversity is *necessary but not sufficient*.

Making survival itself depend on decoding non-kin does not help either. Across three heterogeneous interventions (Figure 1, panel b: C2X2 at 0.25 non-kin stays viable (alive 0.16) yet CF 0.00156; C2X3 at 0.5 non-kin alive 0.10, CF 0.00192; C2X at 100% non-kin crashes to alive 0.046), CF never rose above chance; where the pressure bit, it crashed survival. Non-kin codes are mutually undecodable, so forced cross-listening is acting on noise, which starves foraging: a *pressure-vs-survival tension*. All cross-lineage arms are n=8 (positive-or-inconclusive tier), so we report that a public code *did not* bootstrap under the regimes tested, never that none *can*.

A bidirectional self-audit

Each claim must survive an architecture-matched control, a fair baseline, $\geq$6-seed bootstrap-CI win-margins, frozen pre-registration, a no-oracle redline, and code-grounded adversarial review. Its single methodological claim is *bidirectionality*: it catches deflation as readily as inflation. Two in-scope catches show this: a pilot/formal false *negative* (a formal arm compared against a pilot control had buried a real effect, with generation count the load-bearing knob), and an RNG-offset bug in the headline control whose fix left the conclusion intact and slightly *strengthened* the margin ($+0.045 \to +0.0548$).

Contribution and scope

This separates three claims often conflated in emergent-language work: a decodable code can *evolve*, it can *matter* for survival, and it can still *fail to become public*. A decodable code grows from scratch under survival selection but stays kin-private; selection *accelerates* a clonal-descent channel rather than originating a public one; and no public code bootstraps even under sustained diversity and direct cross-lineage pressure. The results are scoped to one hand-coded world, one architecture, and per-stage timescales (28–48 generations). We read the wall as motivating a richer world and a consequence-driven learner rather than more selection pressure. All code and seeded verdict JSONs are committed and reproducible.

References

Foerster, J., et al. (2016). Learning to communicate with deep multi-agent reinforcement learning. *NeurIPS*.

Hamilton, W. D. (1964). The genetical evolution of social behaviour. *J. Theor. Biol.*, 7(1).

Kirby, S., Cornish, H., and Smith, K. (2008). Cumulative cultural evolution in the laboratory. *PNAS*, 105(31).

Lazaridou, A., Peysakhovich, A., and Baroni, M. (2017). Multi-agent cooperation and the emergence of (natural) language. *ICLR*.

Ren, Y., et al. (2020). Compositional languages emerge in a neural iterated learning model. *ICLR*.